

Status	Beendet
Begonnen	Sonntag, 9. November 2025, 12:27
Abgeschlossen	Sonntag, 9. November 2025, 13:34
Dauer	1 Stunde 6 Minuten
Punkte	19,00/20,00
Bewertung	9,50 von 10,00 (95%)

Frage 1

Richtig

Erreichte Punkte 1,00 von 1,00

Which level of the memory hierarchy can be accessed the slowest?

Wählen Sie eine Antwort:

☐ Hard disk

☐ Main memory

☒ Tertiary storage

☐ Cache

Your answer is correct.

Die richtige Antwort ist: Tertiary storage

Frage 2

Richtig

Erreichte Punkte 1,00 von 1,00

What is temporal locality?

True	False		
☐	☒	It is the time to retrieve the data from the cache.	☒
☒	☐	It means that the data that is in use is likely to be reused in the near future.	☒
☐	☒	It means that the data that is in use is not likely to be reused in the near future.	☒
☐	☒	It is the time to the access a data item locally in main memory.	☒

It is the time to retrieve the data from the cache.: False

It means that the data that is in use is likely to be reused in the near future.: True

It means that the data that is in use is not likely to be reused in the near future.: False

It is the time to the access a data item locally in main memory.: False

Frage **3**

Richtig

Erreichte Punkte 1,00 von 1,00

Which level of the memory hierarchy can typically store the most data?

Wählen Sie eine Antwort:

- ☒ Tertiary storage ✓
- ☐ Main memory
- ☐ Hard disk
- ☐ Cache

Your answer is correct.

Die richtige Antwort ist: Tertiary storage

Frage **4**

Richtig

Erreichte Punkte 1,00 von 1,00

RAID 0 improves the protection against disk crashes.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **5**

Falsch

Erreichte Punkte 0,00 von 1,00

Row stores have high tuple reconstruction costs.

- ☒ Wahr ✗
- ☐ Falsch

Die richtige Antwort ist 'Falsch'.

Frage 6

Richtig

Erreichte Punkte 1,00 von 1,00

Disk access of analytical queries is reduced when using column stores.

- ☒ Wahr ✓
- ☐ Falsch

Analytical queries typically scan large portions of only a single or few columns, which benefits from the columnar layout with respect to disk IO.

Die richtige Antwort ist 'Wahr'.

Frage 7

Richtig

Erreichte Punkte 1,00 von 1,00

Column stores, in general, yield better performance when doing point queries (e.g., reading a single tuple) compared to row stores.

- ☐ Wahr
- ☒ Falsch ✓

Reconstructing single tuples is especially slow in column stores because tuples are stored in a decomposed fashion (i.e., column by column).

Die richtige Antwort ist 'Falsch'.

Frage 8

Richtig

Erreichte Punkte 1,00 von 1,00

Which of the following statements is true for "fractured mirroring"?

Wählen Sie eine Antwort:

- ☐ The data is mirrored using the RAID technique for replicating data for enhanced fault tolerance in case of broken disks.
- ☐ The data is replicated, twice in row layout (NSM) and twice in column layout (DSM) format, with the goal of handling disk failures.
- ☒ The data is stored twice, once in row layout (NSM) and once in column layout (DSM) format to adaptively pick the suitable layout depending on the workload. ✓
- ☐ The data is stored once, but some attributes are stored in row layout (NSM) and some in column layout (DSM).

Your answer is correct.

Die richtige Antwort ist: The data is stored twice, once in row layout (NSM) and once in column layout (DSM) format to adaptively pick the suitable layout depending on the workload.

Frage **9**

Richtig

Erreichte Punkte 1,00 von 1,00

Vertical partitioning aims at reducing the I/O cost for every single incoming query.

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.

Frage **10**

Richtig

Erreichte Punkte 1,00 von 1,00

Pointer swizzling and unswizzling are used to...

Wählen Sie eine Antwort:

- ☐ ...save offsets inside blocks
- ☒ ...map block references from the database address space to the virtual memory space ✓

Your answer is correct.

Die richtige Antwort ist: ...map block references from the database address space to the virtual memory space

Frage 11

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following relation R:

employee_id	payment_id	amount	date
1	2	50	12/05/2020
2	4	30	21/04/2021
3	3	20	15/07/2020
4	6	100	30/06/2020
5	4	75	11/02/2020
6	5	90	21/03/2020
7	7	310	10/01/2021
8	7	145	25/10/2020

Assume that:

- Each field in a record consumes 4 bytes (integers as well as the date).
- Block size as well as page size is 32 bytes.
- No extra storage overhead is required for any storage model (i.e., page and block are same sized and contain purely data.)
- For each query in the questions below, the disk head is above the first block to be read.
- No further information about the schema exists, especially no info on indices and constraints.

How many blocks are retrieved from disk for the following query?

```
select * from R
where employee_id = 1
```

N-ary storage model: ✓

Decomposition storage model: ✓

Frage **12**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following relation R:

employee_id	payment_id	amount	date
1	2	50	12/05/2020
2	4	30	21/04/2021
3	3	20	15/07/2020
4	6	100	30/06/2020
5	4	75	11/02/2020
6	5	90	21/03/2020
7	7	310	10/01/2021
8	7	145	25/10/2020

Assume that:

- Each field in a record consumes 4 bytes (integers as well as the date).
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- No extra storage overhead is required for any storage model (i.e., page and block are same sized and contain purely data.)
- For each query in the questions below, the disk head is above the first block to be read.
- No further information about the schema exists, especially no info on indices and constraints.

How many blocks are retrieved from disk for the following query?

```
select employee_id from R
where date between 01/05/2020 and 01/07/2020
```

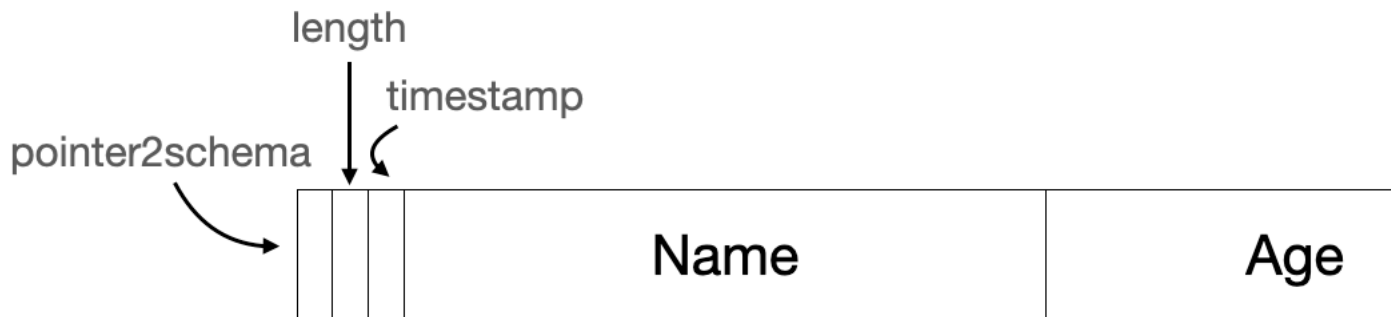
N-ary storage model: ✓Decomposition storage model: ✓

Frage 13

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following record structure:



The fields consume the following space:

Field	Size (bytes)
pointer2schema	1
length	1
timestamp	1
name	28
age	2
state	6

How large is one record if the memory is 4-byte aligned?

Note: all fields are aligned separately.

Antwort: ✓

$4 + 4 + 4 + 28 + 4 + 8$

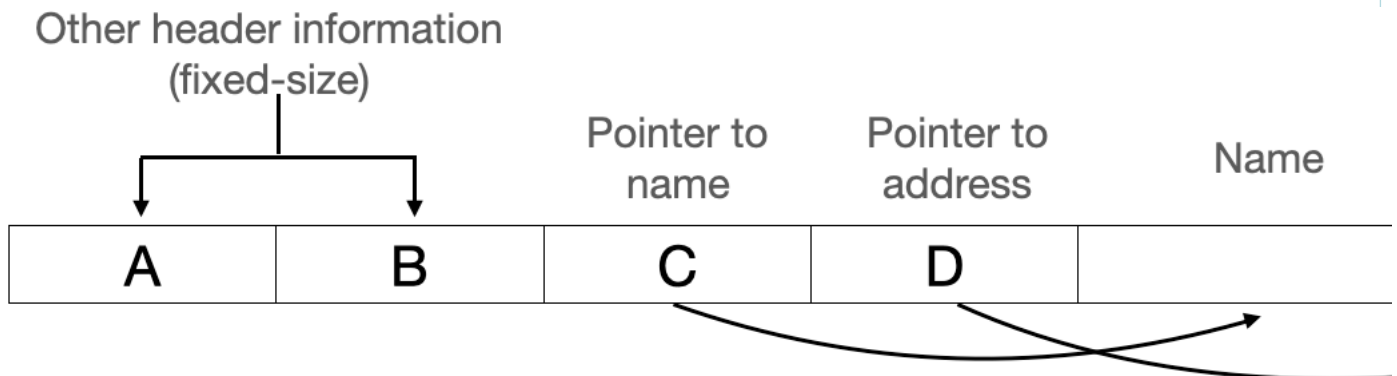
Die richtige Antwort ist: 52

Frage 14

Richtig

Erreichte Punkte 1,00 von 1,00

Assume the following variable length record structure:



A record consists of a **record header and record data**. A record header consists of 4 components: A, B, C, and D. A record data consists of: Name, Address.

The **record header** requires a fixed number of bytes (A and B in the figure above) for book-keeping tasks. It additionally requires C bytes for a pointer to the (variable-sized) **name** field and D bytes for a pointer to the (variable-sized) **address** field. The sizes required by the **header fields of the record** are as follows:

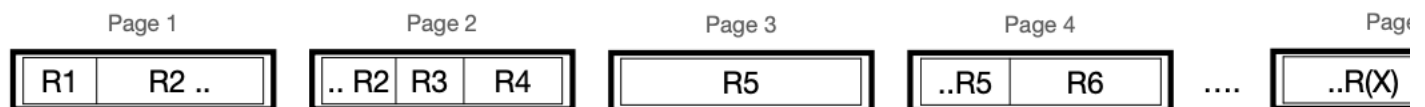
- A = 8 bytes
- B = 4 bytes
- C = 2 bytes
- D = 2 bytes

Consider the following data table. The numbers in parentheses indicate the number of bytes required to store the value.

Name	Address
John Doe (8)	3734 Capitol Avenue (19)
Bruce Wayne (11)	461 Still Street (16)
George Cross (12)	1001 Clover Drive (17)
Johnny Riley (12)	4919 Modoc Alley (16)
Alfred Thaddeus Crane (21)	3895 Owagner Lane (17)
Beryl Hutchinson (16)	4558 Twin Willow Lane (21)

Furthermore, assume that:

- Each variable sized field requires 2 additional bytes (apart from raw storage requirement)
- Each page consists of Page Header and Page Data, i.e multiple records that can be stored on a page.
- Page Header has a fixed size of 12 bytes
- The tuples are stored as row store or NSM
- The records can span multiple blocks, and do not require additional overhead (for example fragment bits etc) apart from the storage requirement for the record
- The records can be placed one after the other without any additional overhead (for example fragment pointers etc.) to maintain as shown below:



How many pages are required if the page size is 128 bytes?

Antwort: ✓

records = [8 + 19, 11 + 16, 12 + 17, 12 + 16, 21 + 17, 16 + 21] sum_records = sum(records) + (8 + 4 + 2 + 2 + 2 + 2) * len(records) sum_records // (128 - 12) + 1

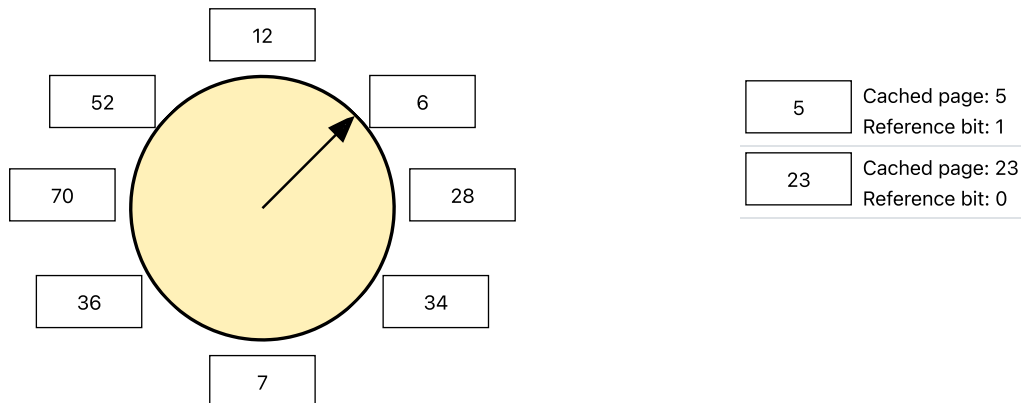
Die richtige Antwort ist: 3

Frage 15

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the CLOCK (a.k.a. Second Chance) algorithm. The system cache has space for 8 pages, and the current content of the cache is shown in the figure below.



Assume that when a page is accessed and it is already in the cache, its reference bit is set to 1 (if it is 0, it is changed to 1; if it is 1, then it is left as 1). The clock hand does not move in such a case (i.e., it stays where it was before this access). The clock hand only moves when the accessed page is not in the cache and the cache is looking for a page to evict by moving clockwise. When a new page is brought into the cache its reference bit is set to 1.

Also assume that in the first figure; red box indicates that the reference bit is set to 1, and orange box indicates that reference bit is set to 0.

Consider the following sequence of page requests to the cache:

20, 15, 36, 6, 42, 3, 61, 89

Fill in the pages of the clock with regards to the page numbers after the last request has been completed (after all the pages in the series are accessed).



Frage 16

Richtig

Erreichte Punkte 1,00 von 1,00

The cache in the system can hold 3 pages in total and is initially empty. Consider the following sequence of page requests to the cache: 15, 4, 6, 15, 10, 11, 6, 9, 12, 1

How many cache hits will occur if the cache replacement policy is FIFO?

Additional information:

Cache Hit - A cache hit occurs when a requested page is **found** in the cache.

Cache Miss - A cache miss occurs when a requested page is **not found** in the cache. Writing on an empty cache is also considered as a cache miss.

Antwort: ✓

Die richtige Antwort ist: 2

Frage 17

Richtig

Erreichte Punkte 1,00 von 1,00

The cache in the system can hold 3 pages in total and is initially empty. Consider the following sequence of page requests to the cache: 15, 4, 6, 15, 9, 11, 6, 12, 15, 6

How many cache misses will occur if the cache replacement policy is FIFO?

Additional information:

Cache Hit - A cache hit occurs when a requested page is **found** in the cache.

Cache Miss - A cache miss occurs when a requested page is **not found** in the cache. Writing on an empty cache is also considered as a cache miss.

Antwort: ✓

Die richtige Antwort ist: 8

Frage 18

Richtig

Erreichte Punkte 1,00 von 1,00

The cache in the system can hold 3 pages in total and is initially empty. Consider the following sequence of page requests to the cache: 15, 4, 6, 15, 9, 11, 6, 9, 12, 1

How many cache hits will occur if the cache replacement policy is LRU?

Additional information:

Cache Hit - A cache hit occurs when a requested page is **found** in the cache.

Cache Miss - A cache miss occurs when a requested page is **not found** in the cache. Writing on an empty cache is also considered as a cache miss.

Antwort: ✓

Die richtige Antwort ist: 2

Frage **19**

Richtig

Erreichte Punkte 1,00 von 1,00

We have a query plan which has a burst access pattern, i.e., it accesses a particular sections of data at a time once, and then never needs that data again.

Given this workload, which caching strategy is more efficient?

Wählen Sie eine Antwort:

- ☐ LFU
- ☒ LRU ✓

Your answer is correct.

Die richtige Antwort ist: LRU

Frage **20**

Richtig

Erreichte Punkte 1,00 von 1,00

Consider the following queries A1 and A2:

A1: `SELECT * FROM R WHERE age > 40`

A2: `SELECT name FROM R WHERE age > 25`

Can A1 be answered from the resulting tuples from A2?

- ☐ Wahr
- ☒ Falsch ✓

Die richtige Antwort ist 'Falsch'.